Prototype 1
Plan for Prototype 2

Palmeiras

Prototype 1 (CFP)

Prototype 1 (Background)

- Checked to see whether the 3D designed (PLA) part and the squeegee would actually clean debris from a mirror
- Added and tested various weights on the prototype to see how much force was necessary to clean it
- Successfully cleaned the mirror, 543 grams working the best

Prototype 1 (Issues/Questions)

- The amount of weight may not be enough to clean in one pass.
- Where do we store the dirt/anything the squeegee has picked up?
- Do we need a way to clean the squeegee as well? How do we do that?

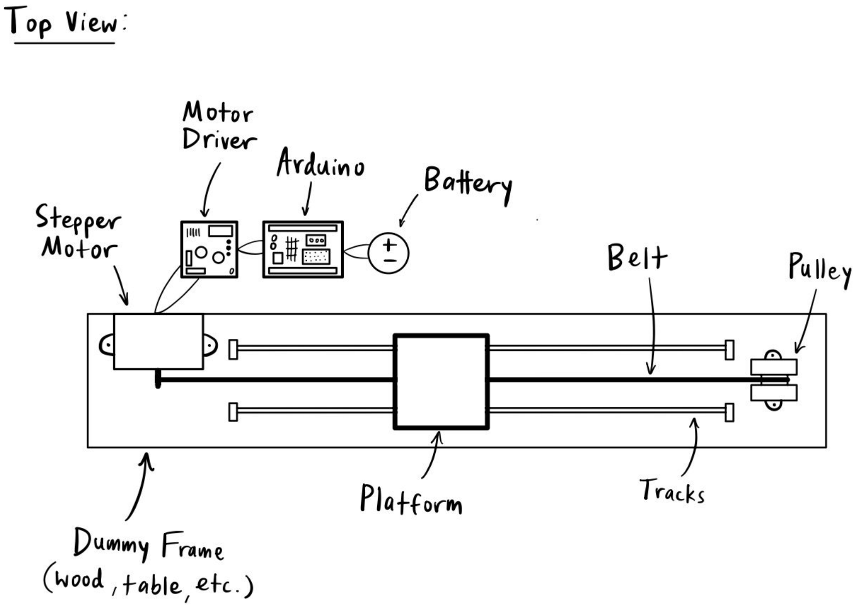
Prototype 2 (Define)

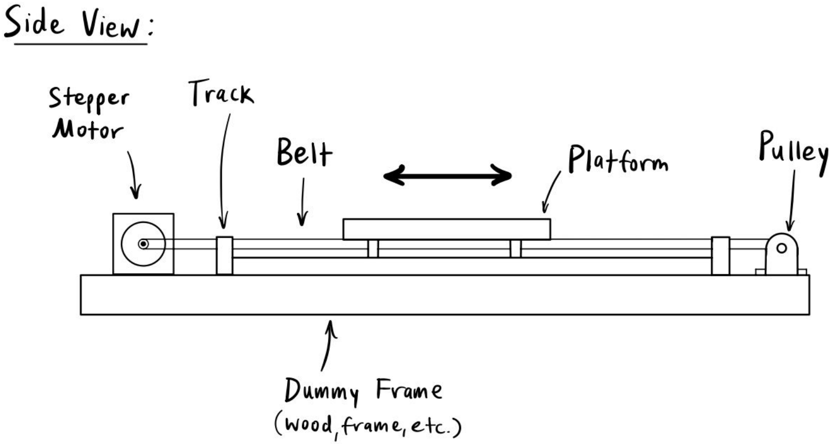
The second prototype will be attempting to move in the x direction

Necessary Items for Prototype 2:

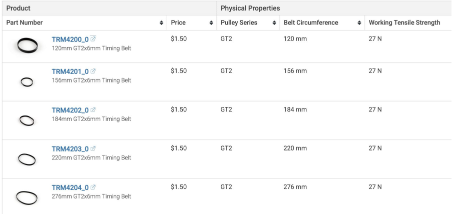
1. 2 motors/drivers (minimum) (only need one for prototype 2)
2. 1 controller
3. Belts for the pulley

Prototype 2 (Sketch)



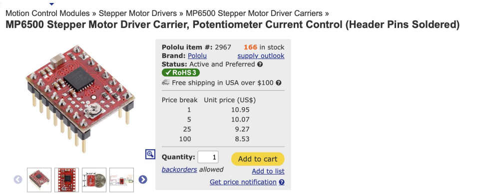


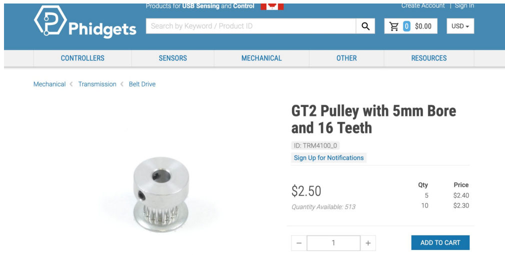
Prototype 2 (Ideate)

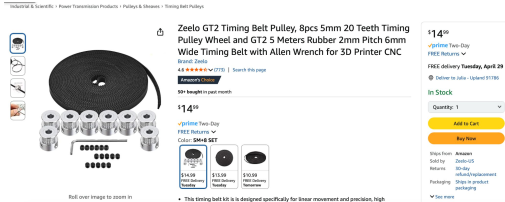


Part Number	Price	Pulley Series	Belt Circumference	Working Tensile Strength
TRM4200_0 120mm GT2x6mm Timing Belt	$1.50	GT2	120 mm	27 N
TRM4201_0 156mm GT2x6mm Timing Belt	$1.50	GT2	156 mm	27 N
TRM4202_0 184mm GT2x6mm Timing Belt	$1.50	GT2	184 mm	27 N
TRM4203_0 220mm GT2x6mm Timing Belt	$1.50	GT2	220 mm	27 N
TRM4204_0 276mm GT2x6mm Timing Belt	$1.50	GT2	276 mm	27 N

- Will be attached to a pulley system, where the squeegee is attached, and moving left and right
- Need one motor (stepper)
- Belts to connect motor and pulley, a track system for the squeegee to move on
- A controller (Arduino) for the system







Prototype 2 (Calculate)

- Capstone Inventory Stepper Motor
  - Max Torque (12V DC) of 0.1467 Nm
- With a shaft diameter of 5mm, we can expect to be able to move ~58.68 N or ~6 kg
- Need 543 grams (~5.32 N) applied to squeegee
- This leaves ~5.5 kg of mass to build the cleaning contraption
- Stepper motor weighs 0.21 kg

$$0.1467 \ N\cdot m = F\,(0.0025 \ m)$$

$$F = 58.68 \ N$$

$$\Rightarrow \quad m = \frac{F}{g} = \frac{56.68 \ N}{9.81 \ \frac{N}{kg}}$$

$$m = 5.98 \ kg$$

Timeline

Prototype 2: Moves in the x direction

Prototype 3: Moves in the y direction. Water/solution dissemination system

Prototype 4: Build the frame

Prototype 5: Clean it up